

Lecture Notes on Quantum Field Theory II

March 23, 2026

Contents

1	The Dirac Field	2
1.1	Lorentz Invariance	2
1.2	The Dirac Equation	8
1.3	Lorentz covariance of the Dirac equation	13

1 The Dirac Field

1.1 Lorentz Invariance

Readings

- Peskin & Schroeder Sec. 3.1
- Zee Chap. II

Relativity and Quantum Mechanics

In nonrelativistic quantum mechanics, the dynamics of a particle is described by the Schrödinger equation

$$i\frac{\partial}{\partial t}\psi(t, \mathbf{x}) = -\frac{\nabla^2}{2m}\psi(t, \mathbf{x}). \quad (1.1)$$

However this equation is not compatible with special relativity, because time and space appear differently: the time derivative appears linearly while spatial derivatives appear quadratically.

In special relativity space and time must be treated on equal footing. The fundamental symmetry of spacetime is the **Lorentz symmetry**. **Lorentz invariance** means that if the original fields satisfy the equations of motion, the transformed fields must also satisfy the same equations.

Lorentz Transformations

In special relativity we combine time and space into spacetime coordinates

$$x^\mu = (t, \mathbf{x}) \quad (1.2)$$

where $\mu = 0, 1, 2, 3$. The invariant spacetime interval is

$$s^2 = -t^2 + \mathbf{x}^2. \quad (1.3)$$

Lorentz transformations are linear transformations

$$x'^\mu = \Lambda^\mu_\nu x^\nu \quad (1.4)$$

that leave the spacetime interval invariant

$$\eta_{\rho\sigma} \Lambda^\rho_\mu \Lambda^\sigma_\nu = \eta_{\mu\nu}, \quad (1.5)$$

where the Minkowski metric in our convention is

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (1.6)$$

For example, a boost in the x direction is

$$t' = \gamma(t - vx), \quad (1.7)$$

$$x' = \gamma(x - vt), \quad (1.8)$$

where $\gamma = 1/\sqrt{1 - v^2}$.

Relativistic Wave Equations

To construct relativistic wave equations we use Lorentz invariant operators. The derivative operator is

$$\partial_\mu = \left(\frac{\partial}{\partial t}, \nabla \right). \quad (1.9)$$

Using the Minkowski metric we define

$$\partial^\mu = \eta^{\mu\nu} \partial_\nu. \quad (1.10)$$

In our metric convention $(-, +, +, +)$ we obtain

$$\partial_\mu \partial^\mu = -\frac{\partial^2}{\partial t^2} + \nabla^2. \quad (1.11)$$

Using this operator we obtain the Klein-Gordon equation

$$(\partial_\mu \partial^\mu - m^2)\phi = 0. \quad (1.12)$$

This equation is Lorentz invariant and describes a relativistic scalar particle.

Limitations of Relativistic Quantum Mechanics

One may attempt to construct a relativistic quantum theory by writing wave equations for a single particle, such as the Klein-Gordon equation or the Dirac equation. However, this approach faces fundamental difficulties.

- First, in relativistic physics the number of particles is not conserved. Processes such as particle creation and annihilation can occur when the energy is sufficiently large. For example, an electron–positron pair can be created from a photon. A single-particle wave function $\psi(\mathbf{x}, t)$ cannot describe processes in which the number of particles changes.
- Second, the relativistic wave equations introduce negative-energy solutions. Interpreting these states within single-particle quantum mechanics leads to conceptual problems.

These difficulties indicate that relativistic quantum mechanics cannot be consistently formulated as a theory of a fixed number of particles. Instead, the correct framework is quantum field theory, in which fields are promoted to operators defined at each spacetime point and particles are interpreted as excitations of these fields.

We refer the reader to the discussion in Quantum Field Theory I for a more detailed explanation of these issues.

Transformation of Fields

The discussion above indicates that relativistic quantum theory must be formulated in terms of fields rather than single-particle wave functions. A field is a physical quantity defined at every spacetime point,

$$\phi(x), \quad x^\mu = (t, \mathbf{x}).$$

Since spacetime coordinates transform under Lorentz transformations, fields must transform in a consistent way so that the physical laws remain Lorentz invariant. In the following we discuss how different types of fields transform under Lorentz transformations.

Under a Lorentz transformation the coordinates change as

$$x \rightarrow x' = \Lambda x. \quad (1.13)$$

Fields transform according to their representation of the Lorentz group.

Scalar field (spin 0) A scalar field is defined as a field whose value is invariant under Lorentz transformations. This means that the value of the field at a given spacetime event does not depend on the choice of inertial frame. For a scalar field we require

$$\phi'(x') = \phi(x). \quad (1.14)$$

This equation states that the value of the field at the same spacetime event is the same in both coordinate systems. Since

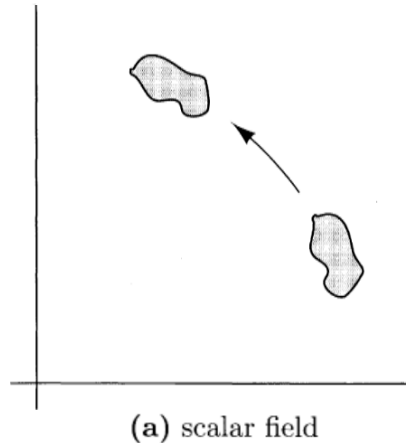
$$x = \Lambda^{-1}x',$$

we can rewrite the transformation law as

$$\phi'(x') = \phi(\Lambda^{-1}x'). \quad (1.15)$$

Finally, renaming the coordinate $x' \rightarrow x$, we obtain the commonly used form

$$\boxed{\phi(x) \rightarrow \phi'(x) = \phi(\Lambda^{-1}x)} \quad (1.16)$$



Example: Lorentz invariance of the Klein–Gordon Lagrangian Consider the Klein–Gordon Lagrangian density

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi + \frac{1}{2}m^2\phi^2. \quad (1.17)$$

Using the scalar field transformation rule

$$\phi \rightarrow \phi'(x) = \phi(\Lambda^{-1}x), \quad (1.18)$$

we can determine how the derivatives transform. Using the chain rule we obtain

$$\partial_\mu\phi'(x) = \frac{\partial}{\partial x^\mu}\phi(\Lambda^{-1}x) = (\Lambda^{-1})^\nu{}_\mu\partial_\nu\phi(\Lambda^{-1}x). \quad (1.19)$$

Therefore

$$\partial_\mu\phi'(x)\partial^\mu\phi'(x) = (\Lambda^{-1})^\nu{}_\mu(\Lambda^{-1})^\rho{}_\sigma\eta^{\mu\sigma}\partial_\nu\phi(\Lambda^{-1}x)\partial_\rho\phi(\Lambda^{-1}x). \quad (1.20)$$

Using the property of Lorentz transformations

$$\eta_{\mu\nu}\Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma = \eta_{\rho\sigma}, \quad (1.21)$$

one finds

$$\partial_\mu\phi'(x)\partial^\mu\phi'(x) = \partial_\mu\phi(\Lambda^{-1}x)\partial^\mu\phi(\Lambda^{-1}x). \quad (1.22)$$

Similarly,

$$\phi'^2(x) = \phi^2(\Lambda^{-1}x). \quad (1.23)$$

Therefore the Lagrangian density transforms as

$$\boxed{\mathcal{L}(x) \rightarrow \mathcal{L}'(x) = \mathcal{L}(\Lambda^{-1}x)}. \quad (1.24)$$

This means that the Lagrangian density is a Lorentz scalar. The action is

$$S = \int d^4x \mathcal{L}(x). \quad (1.25)$$

Since Lorentz transformations satisfy

$$d^4x = d^4x', \quad (1.26)$$

the action is invariant,

$$S' = S. \quad (1.27)$$

Because the equations of motion follow from the stationary condition

$$\delta S = 0, \quad (1.28)$$

the resulting Euler–Lagrange equation

$$(\partial_\mu\partial^\mu - m^2)\phi = 0 \quad (1.29)$$

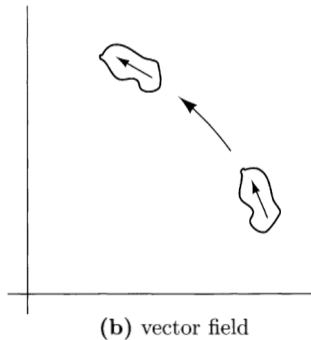
is automatically Lorentz invariant. Therefore, **if the Lagrangian density is a Lorentz scalar, the equations of motion derived from it are automatically Lorentz invariant.**

Vector field (spin 1) To understand how a vector field transforms, it is useful to first consider ordinary spatial rotations. Let $V^i(x)$ be a vector field in three-dimensional space. Under a rotation R , the coordinates transform as

$$x \rightarrow x' = Rx.$$

The vector field transforms as

$$V^i(x) \rightarrow V'^i(x) = R^i{}_j V^j(R^{-1}x). \quad (1.30)$$



This transformation has two effects.

- The argument of the field changes from x to $R^{-1}x$ because the spacetime point itself is transformed.
- The components of the vector rotate according to the rotation matrix R^i_j .

In other words, a rotation affects both the position at which the field is evaluated and the orientation of the vector itself.

The relativistic generalization of this transformation is obtained by replacing rotations with Lorentz transformations. For a four-vector field $V^\mu(x)$ we therefore have

$$\boxed{V^\mu(x) \rightarrow V'^\mu(x) = \Lambda^\mu_\nu V^\nu(\Lambda^{-1}x)}. \quad (1.31)$$

Tensor field The transformation rule for vector fields can be generalized to tensor fields. Consider a rank-two tensor field $T^{\mu\nu}(x)$. Under a Lorentz transformation, the tensor field transforms as

$$\boxed{T^{\mu\nu}(x) \rightarrow T'^{\mu\nu}(x) = \Lambda^\mu_\rho \Lambda^\nu_\sigma T^{\rho\sigma}(\Lambda^{-1}x)}. \quad (1.32)$$

Each Lorentz index introduces one factor of the Lorentz transformation matrix Λ . More generally, for a tensor with multiple indices, each index transforms independently with its own Lorentz matrix.

Index matching and Lorentz covariance Lorentz symmetry imposes a powerful constraint on the form of relativistic equations. If every term in an equation carries the same set of free Lorentz indices, then each term transforms with exactly the same Lorentz matrices. Therefore the equation preserves its form under Lorentz transformations. In this sense, matching Lorentz indices acts as a “machine” that guarantees Lorentz covariance.

E.x. Consider the equation

$$A^\mu + B^\mu = 0. \quad (1.33)$$

Under a Lorentz transformation we obtain

$$A'^\mu = \Lambda^\mu_\nu A^\nu, \quad B'^\mu = \Lambda^\mu_\nu B^\nu.$$

Both terms acquire the same Lorentz matrix, so the equation becomes

$$\Lambda^\mu_\nu A^\nu + \Lambda^\mu_\nu B^\nu = 0, \quad (1.34)$$

which is equivalent to the original equation. Therefore the equation is Lorentz covariant.

However, if the indices do not match, the equation will generally not be Lorentz covariant. For example, consider

$$A^\mu + C = 0, \quad (1.35)$$

where C is a scalar. Under a Lorentz transformation we obtain

$$A'^\mu = \Lambda^\mu_\nu A^\nu, \quad C' = C.$$

The two terms now transform differently, so the transformed equation becomes

$$\Lambda^\mu_\nu A^\nu + C = 0, \quad (1.36)$$

which is not equivalent to the original equation. Therefore the equation is not Lorentz covariant.

Spinor field (spin- $\frac{1}{2}$) Scalar, vector and tensor fields transform with powers of the Lorentz matrix Λ . However, there exist other fields whose transformation properties cannot be written in this way. These are called **spinor fields**.

A spinor field transforms as

$$\boxed{\psi_a(x) \rightarrow \psi'_a(x) = S_{ab}(\Lambda) \psi_b(\Lambda^{-1}x)} \quad (1.37)$$

The matrix $S(\Lambda)$ is a representation of the Lorentz group acting on the spinor indices. Unlike vectors, this matrix is not equal to the Lorentz matrix Λ itself. Instead it provides a different representation of the same symmetry.

Continuous symmetry groups The Lorentz symmetry belongs to a class of symmetries called **Lie groups**. A Lie group is a group of transformations that depend smoothly on continuous parameters.

Examples include

- rotations (parameter: angle θ)
- Lorentz transformations (parameters: rotation angles and boosts)
- translations (parameter: displacement a)

Representations A representation of a symmetry group is a way to realize the symmetry transformation using matrices acting on vectors.

For example, a rotation in the plane can be represented by the matrix

$$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}. \quad (1.38)$$

Different physical objects transform according to different representations of the same symmetry group.

- scalar representation
- vector representation
- spinor representation

Generators Continuous symmetry transformations can be built from infinitesimal transformations. For example, a small rotation by an angle $d\theta$ can be written as

$$R(d\theta) = I - i d\theta J. \quad (1.39)$$

The matrix J is called the generator of the rotation.

Then, a finite transformation is obtained by composing infinitely many such infinitesimal transformations. This leads to the exponential form

$$R(\theta) = e^{-i\theta J}. \quad (1.40)$$

Thus, once the generator is known, the full continuous transformation is obtained by exponentiation.

Lie algebra The set of generators together with their commutation relations is called the **Lie algebra** of the group. The commutation relations tell us how different infinitesimal transformations combine.

For example, for spatial rotations the generators J_i satisfy the commutation relations

$$[J_i, J_j] = i\epsilon_{ijk} J_k. \quad (1.41)$$

From algebra to representation To construct a representation of the group, we first find matrices that represent the generators and satisfy the same commutation relations. Once such matrices are known, the corresponding finite transformations are obtained by exponentiation.

For example, for spin- $\frac{1}{2}$ particles we may choose

$$J_i = \frac{\sigma_i}{2}, \quad (1.42)$$

where σ_i are the Pauli matrices. Since these matrices satisfy the rotation algebra, they provide a representation of the generators. The corresponding finite rotation is then

$$U(\theta) = e^{-i\theta_i J_i} = e^{-i\theta_i \sigma_i / 2}. \quad (1.43)$$

This gives a 2×2 representation of rotations acting on spinors.

Generators of the Lorentz group The Lorentz group is also a Lie group, so its finite transformations are obtained by exponentiating its generators.

To construct the generators, it is useful to first recall the generators of spatial rotations. In three dimensions, the generators of rotations can be written as

$$J_i = i\epsilon_{ijk}x^j\partial_k. \quad (1.44)$$

These generators satisfy the rotation algebra

$$[J_i, J_j] = i\epsilon_{ij}{}^k J_k. \quad (1.45)$$

To generalize this to relativistic spacetime, we introduce antisymmetric generators

$$J^{\mu\nu} = i(x^\mu\partial^\nu - x^\nu\partial^\mu). \quad (1.46)$$

These are the generators of Lorentz transformations. Because the indices are antisymmetric,

$$J^{\mu\nu} = -J^{\nu\mu}, \quad (1.47)$$

there are six independent generators. They correspond to

- three spatial rotations
- three Lorentz boosts.

Lorentz algebra The generators satisfy the Lorentz Lie algebra

$$[J^{\mu\nu}, J^{\rho\sigma}] = i(g^{\nu\rho}J^{\mu\sigma} - g^{\mu\rho}J^{\nu\sigma} - g^{\nu\sigma}J^{\mu\rho} + g^{\mu\sigma}J^{\nu\rho}). \quad (1.48)$$

Any matrices that represent Lorentz transformations must satisfy the same commutation relations. Therefore, any representation $M(\Lambda)$ of the Lorentz group must be constructed from matrices that obey this algebra.

Example: vector representation For a four-vector V^μ , the generators are represented by 4×4 matrices

$$(J^{\mu\nu})^\alpha{}_\beta = i\left(\delta^\mu_\beta g^{\nu\alpha} - \delta^\nu_\beta g^{\mu\alpha}\right). \quad (1.49)$$

An infinitesimal Lorentz transformation can be written as

$$V^\alpha \rightarrow \left(\delta^\alpha_\beta - \frac{i}{2}\omega_{\mu\nu}(J^{\mu\nu})^\alpha{}_\beta\right)V^\beta, \quad (1.50)$$

where $\omega_{\mu\nu}$ are small antisymmetric parameters.

Examples Different choices of the parameters $\omega_{\mu\nu}$ correspond to different Lorentz transformations.

Rotation example For a rotation in the x - y plane we take

$$\omega_{12} = -\omega_{21} = \theta. \quad (1.51)$$

The corresponding transformation acts on a vector as

$$V \rightarrow \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -\theta & 0 \\ 0 & \theta & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} V. \quad (1.52)$$

Boost example For a boost in the x -direction we take

$$\omega_{01} = -\omega_{10} = \beta. \quad (1.53)$$

The transformation becomes

$$V \rightarrow \begin{pmatrix} 1 & \beta & 0 & 0 \\ \beta & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} V. \quad (1.54)$$

Example: spinor representation For a Dirac spinor $\psi_a(x)$, the generators of Lorentz transformations are represented by 4×4 matrices

$$S^{\mu\nu} = \frac{i}{4} [\gamma^\mu, \gamma^\nu], \quad (1.55)$$

where γ^μ are the Dirac gamma matrices. These matrices satisfy the same Lorentz algebra as the generators $J^{\mu\nu}$ and therefore provide a representation of the Lorentz group acting on spinors. We will introduce more details about the spinor representations in the Dirac fields.

1.2 The Dirac Equation

Readings

- Peskin & Schroeder Sec. 3.2
- Zee Chap. II

Historical motivation

Shortly after the development of quantum mechanics by Schrödinger and Heisenberg, physicists attempted to construct a relativistic wave equation for particles. The first successful relativistic equation was the Klein–Gordon equation,

$$(\partial_\mu \partial^\mu - m^2)\phi = 0, \quad (1.56)$$

which we discussed previously. Although this equation is Lorentz invariant, it is not satisfactory as a single-particle relativistic wave equation.

Two major difficulties arise if the Klein–Gordon equation is interpreted as a wave equation.

- The probability density obtained from the Klein–Gordon current is not positive definite.
- The equation admits negative-energy solutions.

At the time these problems were viewed as technical difficulties rather than indications of a deeper issue. In 1928 Dirac attempted to construct an alternative relativistic wave equation that would overcome these problems. Dirac’s idea was to construct an equation that is

- first order in time derivatives, like the Schrödinger equation,
- Lorentz covariant,
- and reproduces the relativistic dispersion relation

$$p^\mu p_\mu = -m^2. \quad (1.57)$$

This led Dirac to propose what is now known as the **Dirac equation**, which provides a relativistic description of spin- $\frac{1}{2}$ particles such as the electron.

Dirac’s construction

Dirac began with an equation similar in form to the Schrödinger equation

$$i\partial_t \psi = H\psi, \quad (1.58)$$

where ψ is now a multi-component wavefunction.

Lorentz covariance requires that time and space derivatives appear in a symmetric way. Therefore the Hamiltonian must be linear in spatial derivatives. The most general form is

$$H = \vec{\alpha} \cdot (-i\nabla) + m\beta, \quad (1.59)$$

where $\vec{\alpha}$ and β are constant matrices.

If $\vec{\alpha}$ and β were ordinary numbers, this equation would not be rotationally invariant¹. Dirac’s crucial insight was to interpret α_i and β as matrices acting on a multi-component wave function.

¹If α_i were numbers, the operator $\alpha_i \partial_i$ would contain a fixed spatial vector $\vec{\alpha}$, which selects a preferred direction and therefore breaks rotational invariance.

Thus the wave function must have several components,

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \vdots \\ \psi_n \end{pmatrix}. \quad (1.60)$$

This object is called a **spinor**. Although ψ is written as a column vector, it is not a spacetime vector. The distinction lies in how the object transforms under Lorentz transformations. A vector transforms with the Lorentz matrix $\Lambda^\mu{}_\nu$, while a spinor transforms with a different matrix $S(\Lambda)$ constructed from the generators of the Lorentz group. Objects that transform in this way are called spinors.

A striking difference is that a 360° spatial rotation leaves vectors unchanged but changes the sign of a spinor. Only after a 720° rotation does a spinor return to its original value. This property is characteristic of particles with spin $1/2$.

Consistency with relativistic dispersion

To ensure that the equation describes relativistic particles, Dirac required that the equation admit plane-wave solutions with the relativistic dispersion relation

$$p^\mu p_\mu = -m^2. \quad (1.61)$$

A simple way to guarantee this property is to require that squaring the first-order equation reproduces the Klein–Gordon equation. Starting from

$$i\partial_t\psi = (\vec{\alpha} \cdot (-i\nabla) + m\beta)\psi, \quad (1.62)$$

we square both sides and obtain

$$-\frac{\partial^2}{\partial t^2}\psi = [\vec{\alpha} \cdot (-i\nabla) + m\beta]^2\psi. \quad (1.63)$$

Expanding the right-hand side gives

$$-\alpha^i\alpha^j\partial_i\partial_j\psi + m^2\beta^2\psi - im(\beta\alpha^i + \alpha^i\beta)\partial_i\psi. \quad (1.64)$$

For this expression to reduce to the Klein–Gordon equation

$$(-\partial_t^2 + \nabla^2 - m^2)\psi = 0, \quad (1.65)$$

the matrices α_i and β must satisfy the conditions

$$\alpha^i\alpha^j + \alpha^j\alpha^i = 0 \quad (i \neq j), \quad (1.66)$$

$$(\alpha^i)^2 = 1, \quad (1.67)$$

$$\beta^2 = 1, \quad (1.68)$$

$$\beta\alpha^i + \alpha^i\beta = 0. \quad (1.69)$$

In addition, since the Hamiltonian must be Hermitian, we require

$$(\alpha^i)^\dagger = \alpha^i, \quad \beta^\dagger = \beta. \quad (1.70)$$

If these relations are satisfied, equation (1.62) automatically admits plane-wave solutions with the relativistic dispersion relation.

We therefore need to find matrices that satisfy these algebraic conditions. It is easy to check that constant numbers cannot satisfy these relations. One can also verify that 2×2 matrices are not sufficient, and 3×3 matrices also fail to satisfy all the conditions simultaneously. The smallest matrices that work are 4×4 matrices.

Therefore the minimal dimension of the matrices is

$$n \geq 4. \quad (1.71)$$

In fact, there exist infinitely many matrices satisfying these conditions.

Example solutions One possible solution is

$$\beta = \begin{pmatrix} 1_{2 \times 2} & 0 \\ 0 & -1_{2 \times 2} \end{pmatrix}, \quad \alpha^i = \begin{pmatrix} 0 & \sigma^i \\ \sigma^i & 0 \end{pmatrix}, \quad (1.72)$$

where σ^i are the Pauli matrices.
Another solution is

$$\beta = \begin{pmatrix} 0 & 1_{2 \times 2} \\ 1_{2 \times 2} & 0 \end{pmatrix}, \quad \alpha^i = \begin{pmatrix} \sigma^i & 0 \\ 0 & -\sigma^i \end{pmatrix}. \quad (1.73)$$

One can verify directly that both of these choices satisfy the required relations.

Since α_i and β are 4×4 matrices, the wavefunction ψ must be a four-component object,

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix}. \quad (1.74)$$

The components ψ_α are complex functions. This new object is called a **spinor**. Later we will see that it describes particles with spin $\frac{1}{2}$.

Gamma matrices

For later convenience it is useful to rewrite the equation in a more symmetric form. Starting from

$$i\partial_t\psi = -i\vec{\alpha} \cdot \nabla\psi + m\beta\psi, \quad (1.75)$$

we multiply both sides by β ,

$$i\beta\partial_t\psi = -i\beta\alpha^i\partial_i\psi + m\psi. \quad (1.76)$$

We now introduce the matrices

$$\gamma^0 = i\beta, \quad \gamma^i = i\beta\alpha^i. \quad (1.77)$$

In terms of these matrices the equation becomes

$$(\gamma^\mu\partial_\mu - m)\psi = 0. \quad (1.78)$$

Because different authors use different conventions for the Minkowski metric, factors of i may appear in different places in the literature. In our convention the equation takes the form above.

Clifford algebra

The relations satisfied by α_i and β can be written more compactly in terms of the gamma matrices. They satisfy

$$\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 0 \quad (\mu \neq \nu), \quad (1.79)$$

together with

$$(\gamma^0)^2 = -1, \quad (\gamma^i)^2 = 1. \quad (1.80)$$

These relations can be written in the compact form

$$\boxed{\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}}. \quad (1.81)$$

Here we use the anticommutator

$$\{A, B\} = AB + BA. \quad (1.82)$$

Equation (1.81) defines the **Clifford algebra**. Any set of matrices γ^μ satisfying this relation is called a representation of the Clifford algebra.

From the above properties of α_i and β we also obtain

$$(\gamma^0)^\dagger = -\gamma^0, \quad (\gamma^i)^\dagger = \gamma^i. \quad (1.83)$$

These relations can also be written in the compact form

$$\boxed{(\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0}. \quad (1.84)$$

Explicit representations of the gamma matrices Using the solutions for α^i and β given earlier, we can construct explicit matrices γ^μ . For example, using the first solution we obtain

$$\gamma^0 = \begin{pmatrix} i\mathbf{1}_{2 \times 2} & 0 \\ 0 & -i\mathbf{1}_{2 \times 2} \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & i\sigma^i \\ -i\sigma^i & 0 \end{pmatrix}, \quad (1.85)$$

where σ^i are the Pauli matrices.

Using the second solution for α and β we obtain another set of gamma matrices

$$\gamma^0 = \begin{pmatrix} 0 & i\mathbf{1}_{2 \times 2} \\ i\mathbf{1}_{2 \times 2} & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & -i\sigma^i \\ i\sigma^i & 0 \end{pmatrix}. \quad (1.86)$$

Both sets of matrices satisfy the Clifford algebra (??). Therefore they represent two different representations of the same algebra.

Remarks

- **Massless case**

Consider the case $m = 0$. In this case the dispersion relation becomes

$$p^2 = 0. \quad (1.87)$$

The Dirac equation reduces to

$$i\partial_t \psi = -i\alpha^i \partial_i \psi. \quad (1.88)$$

The matrices α_i must satisfy

$$\{\alpha_i, \alpha_j\} = 0, \quad \alpha_i^2 = 1. \quad (1.89)$$

These conditions can be satisfied by the Pauli matrices

$$\alpha_i = \sigma_i. \quad (1.90)$$

Therefore in the massless case it is sufficient to use 2×2 matrices, and the wavefunction can be written as a two-component spinor

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}. \quad (1.91)$$

This illustrates an important difference between massive and massless particles. Massive fermions require four components, while massless fermions can be described by two-component spinors.

We will discuss this point in more detail later.

- **Conserved current**

One of Dirac's original motivations was that the Dirac equation admits a conserved current, analogous to the probability current in the Schrödinger equation.

Starting from the Dirac equation

$$(\gamma^\mu \partial_\mu - m)\psi = 0, \quad (1.92)$$

The adjoint equation is

$$0 = (\partial_\mu \psi^\dagger) (\gamma^\mu)^\dagger - m\psi^\dagger = (\partial_\mu \psi^\dagger) \gamma^0 \gamma^\mu \gamma^0 - m\psi^\dagger. \quad (1.93)$$

where we use the relation

$$(\gamma^\mu)^\dagger = \gamma^0 \gamma^\mu \gamma^0. \quad (1.94)$$

Then, multiplying (1.93) from the right by γ^0 gives

$$0 = -(\partial_\mu \psi^\dagger) \gamma^0 \gamma^\mu - m\psi^\dagger \gamma^0, \quad (1.95)$$

we define the Dirac adjoint

$$\bar{\psi} \equiv \psi^\dagger \gamma^0. \quad (1.96)$$

In the end, one finds that the Hermitian conjugate of the Dirac equation implies the adjoint equation

$$(\gamma^\mu \partial_\mu + m)\bar{\psi} = 0. \quad (1.97)$$

Multiplying the Dirac equation from the left by $\bar{\psi}$ gives

$$\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi = 0, \quad (1.98)$$

while multiplying the adjoint equation from the right by ψ gives

$$(\partial_\mu\bar{\psi})\gamma^\mu\psi + m\bar{\psi}\psi = 0. \quad (1.99)$$

Adding these two equations, the mass terms cancel and we obtain

$$(\partial_\mu\bar{\psi})\gamma^\mu\psi + \bar{\psi}\gamma^\mu\partial_\mu\psi = 0. \quad (1.100)$$

This can be written as

$$\partial_\mu(\bar{\psi}\gamma^\mu\psi) = 0. \quad (1.101)$$

Therefore the conserved current is

$$j^\mu = \bar{\psi}\gamma^\mu\psi. \quad (1.102)$$

In our convention $(\gamma^0)^2 = -1$, so

$$j^0 = -\psi^\dagger\psi. \quad (1.103)$$

Equivalently, one may define

$$\tilde{j}^\mu = -\bar{\psi}\gamma^\mu\psi, \quad (1.104)$$

for which

$$\tilde{j}^0 = \psi^\dagger\psi \geq 0. \quad (1.105)$$

At the end of the day, we have the positive definite conserved current

$$\tilde{j}^\mu = -\bar{\psi}\gamma^\mu\psi \quad (1.106)$$

satisfying $\partial_\mu\tilde{j}^\mu = 0$ with positive definite classically $\tilde{j}^0 = \psi^\dagger\psi \geq 0$.

• Equivalence of different gamma matrices

There are infinitely many matrices satisfying the Clifford algebra. What is the meaning of these different solutions?

To understand this, consider the Dirac equation as a matrix equation acting on a four-component spinor ψ . Suppose we perform a change of basis in this spinor space

$$\psi \rightarrow \psi' = B\psi, \quad (1.107)$$

where B is an invertible constant matrix. Insert $\psi = B^{-1}\psi'$ into the original Dirac equation:

$$(\gamma^\mu\partial_\mu - m)B^{-1}\psi' = 0, \quad (1.108)$$

Multiplying equation (1.78) by B gives

$$(\gamma'^\mu\partial_\mu - m)\psi' = 0, \quad (1.109)$$

where

$$\gamma'^\mu = B\gamma^\mu B^{-1}. \quad (1.110)$$

Thus ψ' satisfies the same Dirac equation but with a different set of gamma matrices.

It is straightforward to check that the matrices γ'^μ also satisfy the Clifford algebra. Therefore any two sets of gamma matrices related by a similarity transformation describe the same physical theory. They correspond simply to different choices of basis for the spinor space.

Although these representations are mathematically equivalent, different choices of gamma matrices can be more convenient in different physical situations. For example, the representation introduced earlier

$$\gamma^0 = \begin{pmatrix} i\mathbf{1}_{2\times 2} & 0 \\ 0 & -i\mathbf{1}_{2\times 2} \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & i\sigma^i \\ -i\sigma^i & 0 \end{pmatrix} \quad (1.111)$$

is particularly useful for studying the non-relativistic limit, because it makes the connection with the Schrödinger equation more transparent.

On the other hand, another representation

$$\gamma^0 = \begin{pmatrix} 0 & i\mathbf{1}_{2\times 2} \\ i\mathbf{1}_{2\times 2} & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & -i\sigma^i \\ i\sigma^i & 0 \end{pmatrix} \quad (1.112)$$

is often more convenient in the ultra-relativistic regime.

Therefore different representations of the gamma matrices are used in practice depending on the physical problem being studied. Since all such representations are related by similarity transformations, they lead to the same physical predictions.

1.3 Lorentz covariance of the Dirac equation

We now show that the Dirac equation is Lorentz covariant. Lorentz covariance means that the equation has the same form in every Lorentz frame. Under a Lorentz transformation,

$$x^\mu \rightarrow x'^\mu = \Lambda^\mu{}_\nu x^\nu, \quad (1.113)$$

the transformed field should satisfy the same Dirac equation. Before discussing the Dirac field, let us briefly recall simpler examples.

For a scalar field, the active transformation law is

$$\phi(x) \rightarrow \phi'(x) = \phi(\Lambda^{-1}x). \quad (1.114)$$

Similarly, for a vector field such as the Maxwell potential,

$$A_\mu(x) \rightarrow A'_\mu(x) = \Lambda_\mu{}^\nu A_\nu(\Lambda^{-1}x). \quad (1.115)$$

In this case the spacetime argument changes and the vector index also transforms with a Lorentz matrix. Maxwell's equations then keep the same form in all frames. For example,

$$\partial_\mu F^{\mu\nu} = 0 \quad \Longleftrightarrow \quad \partial'_\mu F'^{\mu\nu} = 0. \quad (1.116)$$

For the Dirac equation, the field ψ is neither a scalar nor a spacetime vector. It is a new object, a spinor. Therefore we must determine how ψ transforms under Lorentz transformations.

Lorentz transformation of the Dirac field

We postulate that under a Lorentz transformation Λ , the Dirac field transforms as

$$\psi(x) \rightarrow \psi'(x) = S(\Lambda)\psi(\Lambda^{-1}x), \quad (1.117)$$

where $S(\Lambda)$ is a 4×4 matrix acting on the spinor indices.

This is the active point of view: we keep the coordinate x fixed, but the transformed field at x is expressed in terms of the old field evaluated at the transformed point $\Lambda^{-1}x$.

Lorentz covariance of the Dirac equation means that whenever $\psi(x)$ satisfies

$$(\gamma^\mu \partial_\mu - m)\psi(x) = 0, \quad (1.118)$$

then the transformed field $\psi'(x)$ must also satisfy

$$(\gamma^\mu \partial_\mu - m)\psi'(x) = 0. \quad (1.119)$$

Here the matrices γ^μ themselves do not transform as spacetime vectors. They are fixed matrices acting in spinor space. The Lorentz transformation acts on the field ψ , not on the gamma matrices as dynamical variables.

Condition for Lorentz covariance

To determine $S(\Lambda)$, we compute the derivative of the transformed field:

$$\partial_\mu \psi'(x) = \partial_\mu [S(\Lambda)\psi(\Lambda^{-1}x)] = S(\Lambda)(\Lambda^{-1})^\nu{}_\mu \partial_\nu \psi(\Lambda^{-1}x), \quad (1.120)$$

since $S(\Lambda)$ is a constant matrix for a global Lorentz transformation. Therefore

$$\gamma^\mu \partial_\mu \psi'(x) = \gamma^\mu S(\Lambda)(\Lambda^{-1})^\nu{}_\mu \partial_\nu \psi(\Lambda^{-1}x). \quad (1.121)$$

For (1.119) to follow from the original Dirac equation, it is sufficient that

$$\gamma^\mu S(\Lambda)(\Lambda^{-1})^\nu{}_\mu = \gamma^\nu. \quad (1.122)$$

Multiplying this equation from the left by $S^{-1}(\Lambda)$ gives the standard condition

$$S^{-1}(\Lambda)\gamma^\mu S(\Lambda) = \Lambda^\mu{}_\nu \gamma^\nu. \quad (1.123)$$

Equivalently,

$$\boxed{S(\Lambda)\gamma^\mu S^{-1}(\Lambda) = (\Lambda^{-1})^\mu{}_\nu \gamma^\nu}. \quad (1.124)$$

So the problem reduces to finding a matrix $S(\Lambda)$ satisfying (1.123).

Infinitesimal Lorentz transformations

As usual, it is much easier to first solve the problem for infinitesimal Lorentz transformations. Let

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu, \quad \omega_{\mu\nu} = -\omega_{\nu\mu}, \quad (1.125)$$

where $\omega^\mu{}_\nu$ is infinitesimal. Then

$$(\Lambda^{-1})^\mu{}_\nu = \delta^\mu{}_\nu - \omega^\mu{}_\nu \quad (1.126)$$

to first order in ω . Since Lorentz transformations form a continuous group, the corresponding spinor transformation must also depend smoothly on the parameters $\omega_{\mu\nu}$. Near the identity transformation we can therefore expand $S(\Lambda)$ linearly in $\omega_{\mu\nu}$,

$$S(\Lambda) = 1 - \frac{i}{2} \omega_{\mu\nu} \Sigma^{\mu\nu} + O(\omega^2),$$

where the matrices $\Sigma^{\mu\nu}$ act in spinor space and are called the generators of Lorentz transformations. Similarly,

$$S^{-1}(\Lambda) = 1 + \frac{i}{2} \omega_{\mu\nu} \Sigma^{\mu\nu} \quad (1.127)$$

to first order in ω . Substituting these expressions into (1.123) and keeping only terms linear in ω , we obtain

$$i[\Sigma^{\lambda\rho}, \gamma^\mu] = \eta^{\lambda\mu} \gamma^\rho - \eta^{\rho\mu} \gamma^\lambda. \quad (1.128)$$

Thus the problem is reduced to finding matrices $\Sigma^{\mu\nu}$ satisfying (1.128).

Generators in spinor space

A solution of (1.128) is

$$\Sigma^{\lambda\rho} = \frac{i}{4} [\gamma^\lambda, \gamma^\rho]. \quad (1.129)$$

One can verify this directly by substituting into (1.128) and repeatedly using the Clifford algebra (1.81). The matrices $\Sigma^{\mu\nu}$ are the generators of Lorentz transformations in spinor space. For a finite Lorentz transformation, the corresponding spinor transformation is obtained by exponentiation:

$$S(\Lambda) = \exp \left[-\frac{i}{2} \omega_{\mu\nu} \Sigma^{\mu\nu} \right]. \quad (1.130)$$

Remarks on the generators

- The matrices Σ^{ij} generate spatial rotations in spinor space:

$$\Sigma^{ij} = \frac{i}{4} [\gamma^i, \gamma^j]. \quad (1.131)$$

Since γ^i and γ^j are Hermitian, we find

$$(\Sigma^{ij})^\dagger = \Sigma^{ij}. \quad (1.132)$$

Therefore the spinor transformation corresponding to a pure rotation,

$$S = \exp \left[-\frac{i}{2} \omega_{ij} \Sigma^{ij} \right], \quad (1.133)$$

is unitary.

- The matrices Σ^{0i} generate boosts in spinor space:

$$\Sigma^{0i} = \frac{i}{4} [\gamma^0, \gamma^i]. \quad (1.134)$$

Since $(\gamma^0)^\dagger = \gamma^0$ and $(\gamma^i)^\dagger = -\gamma^i$, we obtain

$$(\Sigma^{0i})^\dagger = -\Sigma^{0i}. \quad (1.135)$$

So Σ^{0i} is anti-Hermitian, and the corresponding boost transformation

$$S = \exp \left[-\frac{i}{2} \omega_{0i} \Sigma^{0i} \right] \quad (1.136)$$

is not unitary.

- For a general Lorentz transformation containing both rotations and boosts,

$$SS^\dagger \neq 1. \quad (1.137)$$

This fact will be important when constructing Lorentz invariant quantities.

Lorentz invariant bilinears

Since S is not unitary in general, the quantity $\psi^\dagger \psi$ is not Lorentz invariant. Indeed,

$$\psi^\dagger \psi \rightarrow \psi'^\dagger \psi' = \psi^\dagger (\Lambda^{-1} x) S^\dagger S \psi (\Lambda^{-1} x), \quad (1.138)$$

which is not equal to $\psi^\dagger \psi$ unless $S^\dagger S = 1$. So we need to find another scalar built from ψ .

From (1.129), one can show that

$$(\Sigma^{\mu\nu})^\dagger = -\gamma^0 \Sigma^{\mu\nu} \gamma^0. \quad (1.139)$$

Therefore

$$S^\dagger = \exp \left[\frac{i}{2} \omega_{\mu\nu} (\Sigma^{\mu\nu})^\dagger \right] = \exp \left[-\frac{i}{2} \omega_{\mu\nu} \gamma^0 \Sigma^{\mu\nu} \gamma^0 \right] = -\gamma^0 \exp \left[\frac{i}{2} \omega_{\mu\nu} \Sigma^{\mu\nu} \right] \gamma^0. \quad (1.140)$$

Using $(\gamma^0)^2 = -1$, one finds

$$S^\dagger = -\gamma^0 S^{-1} \gamma^0. \quad (1.141)$$

This suggests defining the Dirac adjoint

$$\boxed{\bar{\psi} \equiv \psi^\dagger \gamma^0}. \quad (1.142)$$

Then

$$\bar{\psi} \psi \quad (1.143)$$

is a Lorentz scalar. Indeed,

$$\bar{\psi}'(x) \psi'(x) = \psi'^\dagger(x) \gamma^0 \psi'(x) \quad (1.144)$$

$$= \psi^\dagger (\Lambda^{-1} x) S^\dagger \gamma^0 S \psi (\Lambda^{-1} x) \quad (1.145)$$

$$= \psi^\dagger (\Lambda^{-1} x) \gamma^0 \psi (\Lambda^{-1} x) \quad (1.146)$$

$$= \bar{\psi} (\Lambda^{-1} x) \psi (\Lambda^{-1} x). \quad (1.147)$$

It is also useful to write the transformation of $\bar{\psi}$ itself:

$$\bar{\psi}(x) \rightarrow \bar{\psi}'(x) = \bar{\psi}(\Lambda^{-1} x) S^{-1}. \quad (1.148)$$

Similarly, from the transformation law of ψ one finds

$$\begin{aligned} \gamma^\mu \partial_\mu \psi'(x) &= \gamma^\mu S (\Lambda^{-1})^\nu{}_\mu \partial_\nu \psi (\Lambda^{-1} x) = S (S^{-1} \gamma^\mu S) (\Lambda^{-1})^\nu{}_\mu \partial_\nu \psi (\Lambda^{-1} x) \\ &= S \Lambda^\mu{}_\rho \gamma^\rho (\Lambda^{-1})^\nu{}_\mu \partial_\nu \psi (\Lambda^{-1} x) = S \gamma^\mu \partial_\mu \psi (\Lambda^{-1} x). \end{aligned} \quad (1.149)$$

It is convenient to introduce the slash notation

$$\not{\partial} \equiv \gamma^\mu \partial_\mu. \quad (1.150)$$

Then (1.149) becomes

$$\not{\partial} \psi'(x) = S \not{\partial} \psi (\Lambda^{-1} x). \quad (1.151)$$

One can also show that

$$\bar{\psi} \gamma^\mu \psi \quad (1.152)$$

transforms as a Lorentz vector.

Dirac action

Using the Lorentz invariant quantities constructed above, we can write the Dirac action as

$$S = -i \int d^4 x \bar{\psi} (\gamma^\mu \partial_\mu - m) \psi. \quad (1.153)$$

This action is Lorentz invariant because $\bar{\psi} \psi$ is a scalar and $\bar{\psi} \gamma^\mu \partial_\mu \psi$ is also a scalar. The overall factor of i is included so that the action is real. Indeed,

$$(\bar{\psi} \psi)^\dagger = -\bar{\psi} \psi \quad (1.154)$$

and

$$(\bar{\psi} \gamma^\mu \partial_\mu \psi)^\dagger = -\bar{\psi} \gamma^\mu \partial_\mu \psi + \text{total derivative}. \quad (1.155)$$

The total derivative contributes only a boundary term to the action, which is assumed to vanish at infinity.

The action (1.153) reproduces the Dirac equation. Treating ψ and $\bar{\psi}$ as independent variables, variation with respect to $\bar{\psi}$ gives

$$(\gamma^\mu \partial_\mu - m) \psi = 0, \quad (1.156)$$

while variation with respect to ψ gives the adjoint equation for $\bar{\psi}$:

$$(\gamma^\mu \partial_\mu + m) \bar{\psi} = 0. \quad (1.157)$$